



Institute_{of}
Data

Boosting Developer Productivity with Data- Driven Insights

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Mini Project 2 – Data Science

Project Overview

Problem
Identification



Exploratory
Data Analysis



Results



Assumptions &
Limitations



Stakeholders &
Value
Proposition



Applying
Supervised
Learning
Models

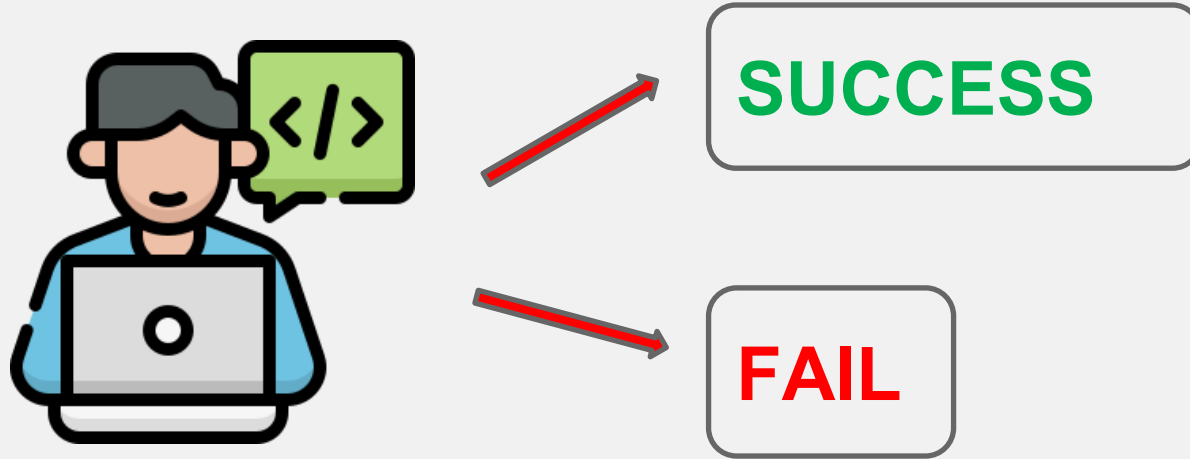


Conclusions &
Recommendations



Problem – PRODUCTIVITY

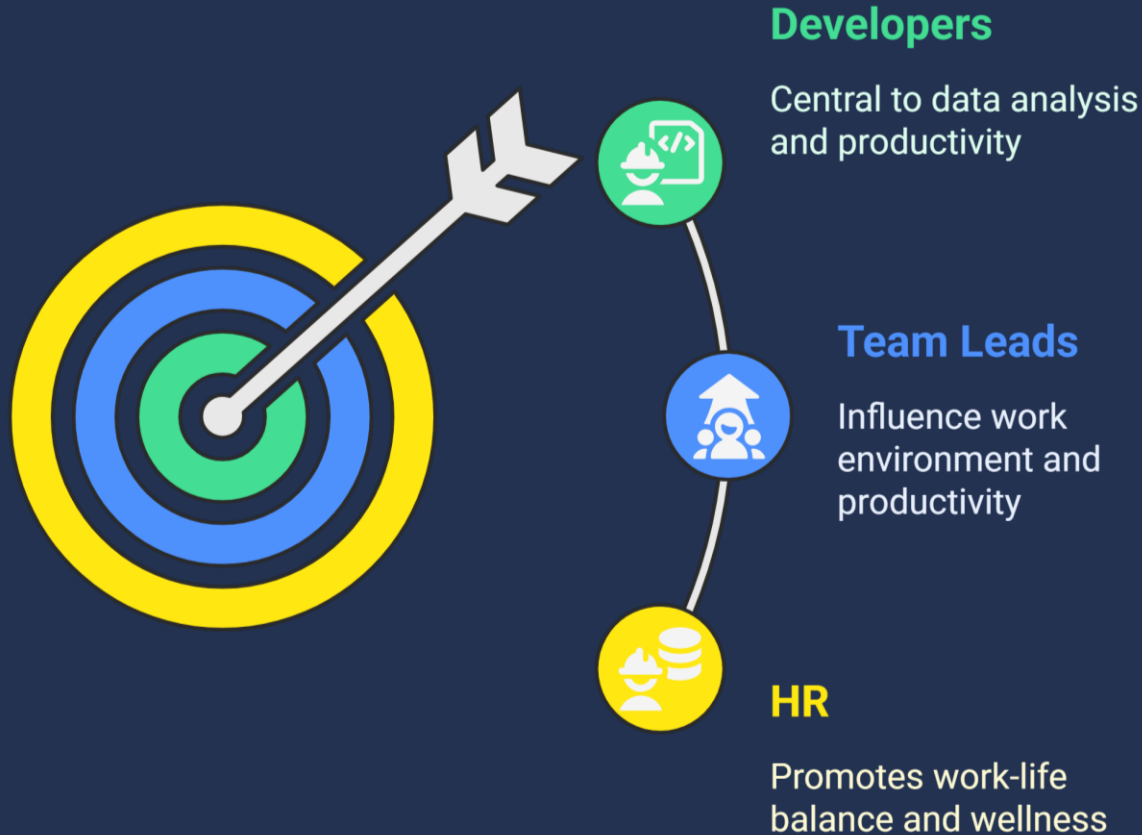
- Distractions: ~ 23 mins to get back on task [1].
- AI Implementation - 50% of developers adopting AI Assisted-development during 2023 (Reported by GitHub) [2].



[1] <https://www.fastcompany.com/944128/worker-interrupted-cost-task-switching>

[2] https://www.gitclear.com/coding_on_copilot_data_shows_ais_downward_pressure_on_code_quality

Who is affected? (Stakeholders)



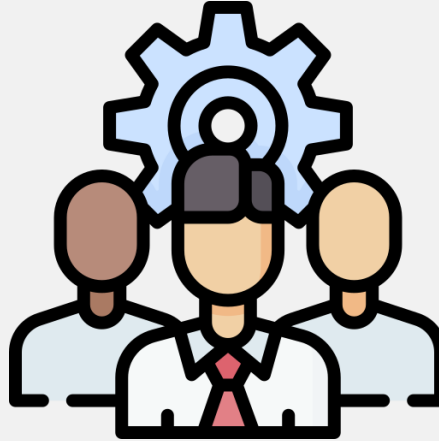
Value Proposition

Developers



Personal **growth**
Efficiency
Career advancement

Team Leads



Proactive **management**
Spot **productivity** patterns
Evidence-based **decisions**

Human Resources



Employee **well-being** & retention
Identify early signs of **burnout**
Justify wellness initiatives

Exploratory Data Analysis (EDA)

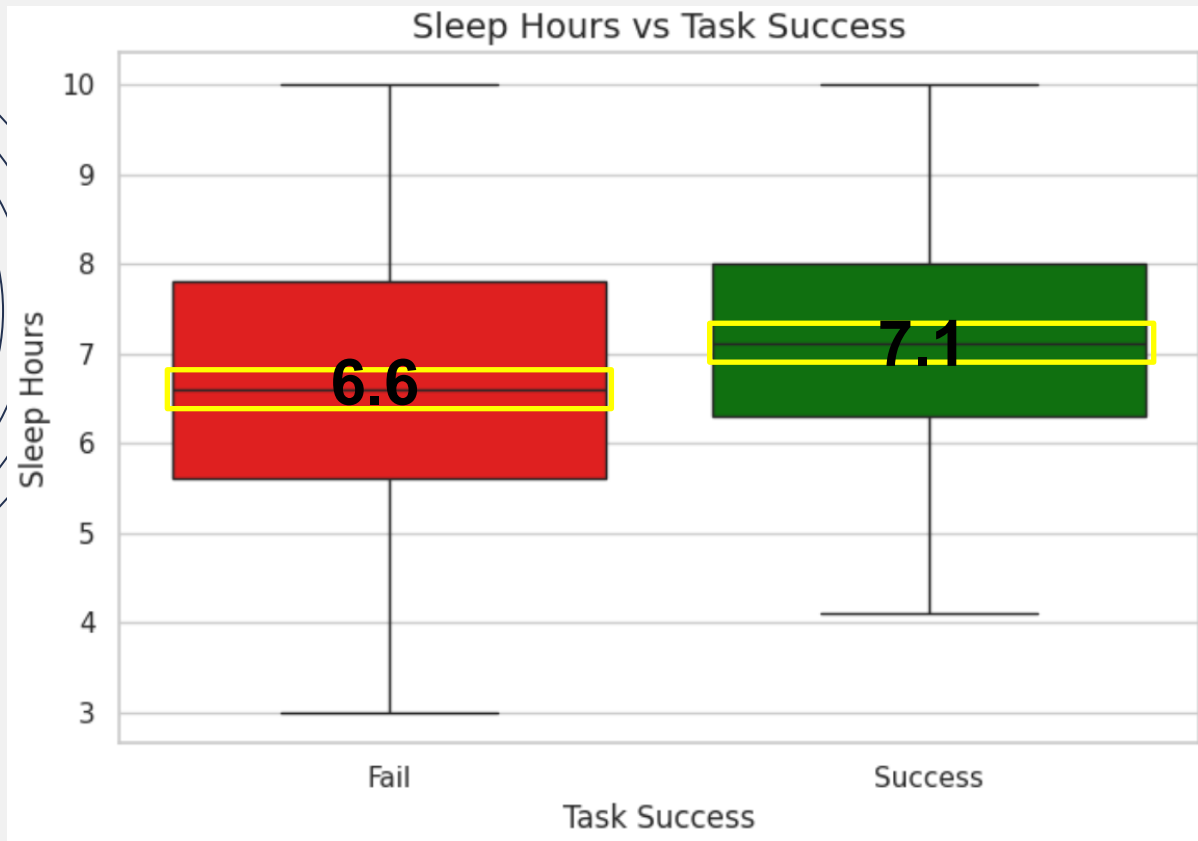


- Sleep Hours
- Coffee Intake
- Distractions
- Hours Coding
- AI Usage Hours
- Bugs Reported
- Commits
- Cognitive Load
- **Task Success** – whether the daily productivity goal was achieved (0/1)

Correlations with Task Success?

'ai_dev_productivity.csv' – Productivity of AI developers
(9 columns, 500 rows)

<https://www.kaggle.com/datasets/atharvasoundankar/ai-developer-productivity-dataset/data>



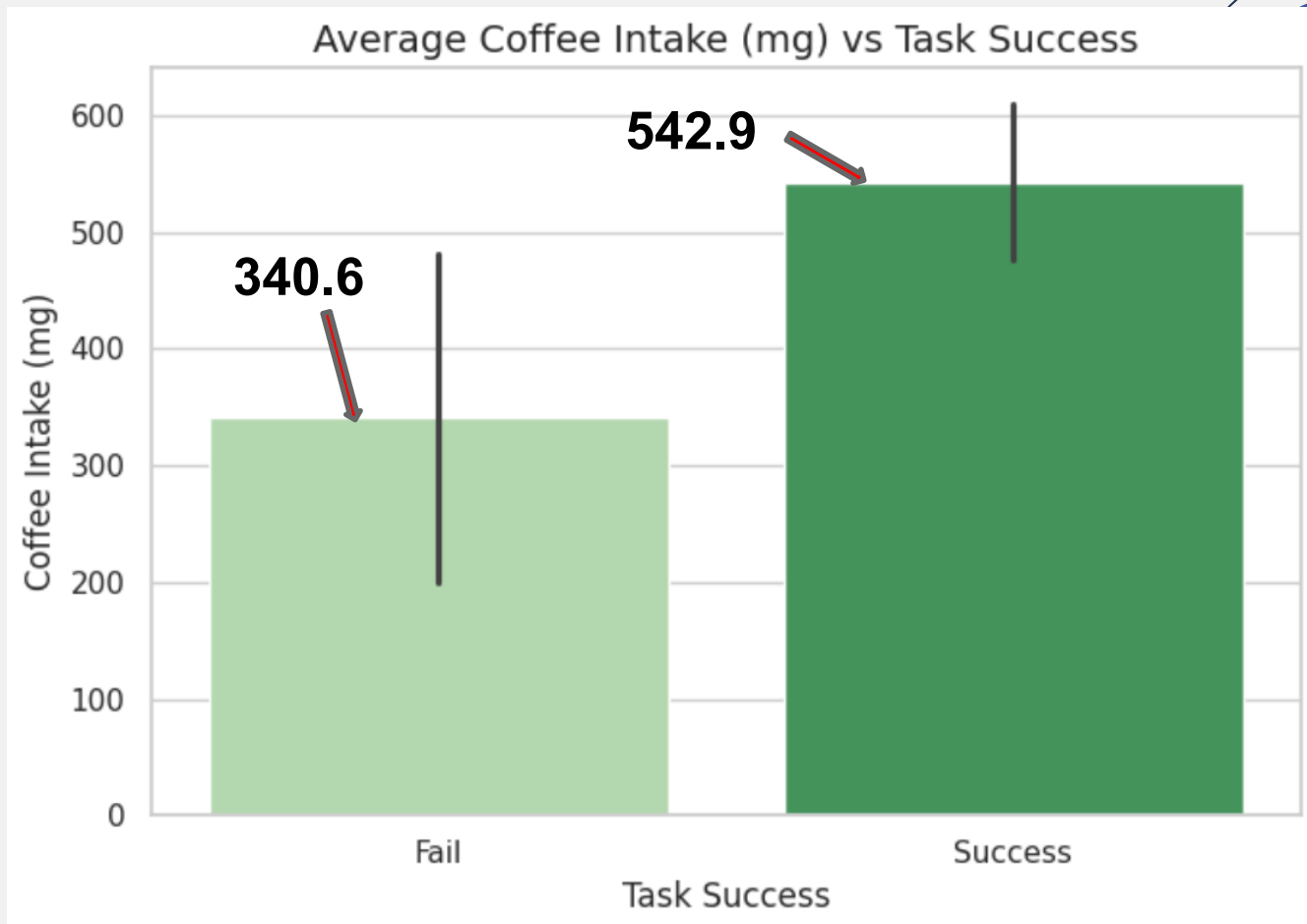
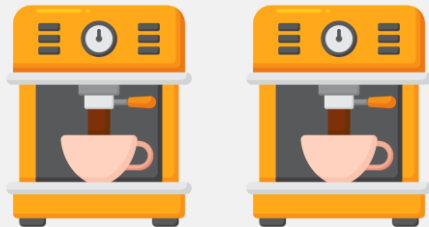
- Sleep Hours
- Coffee Intake
- Distractions
- Hours Coding
- AI Usage Hours
- Task Success - whether the daily productivity goal was achieved



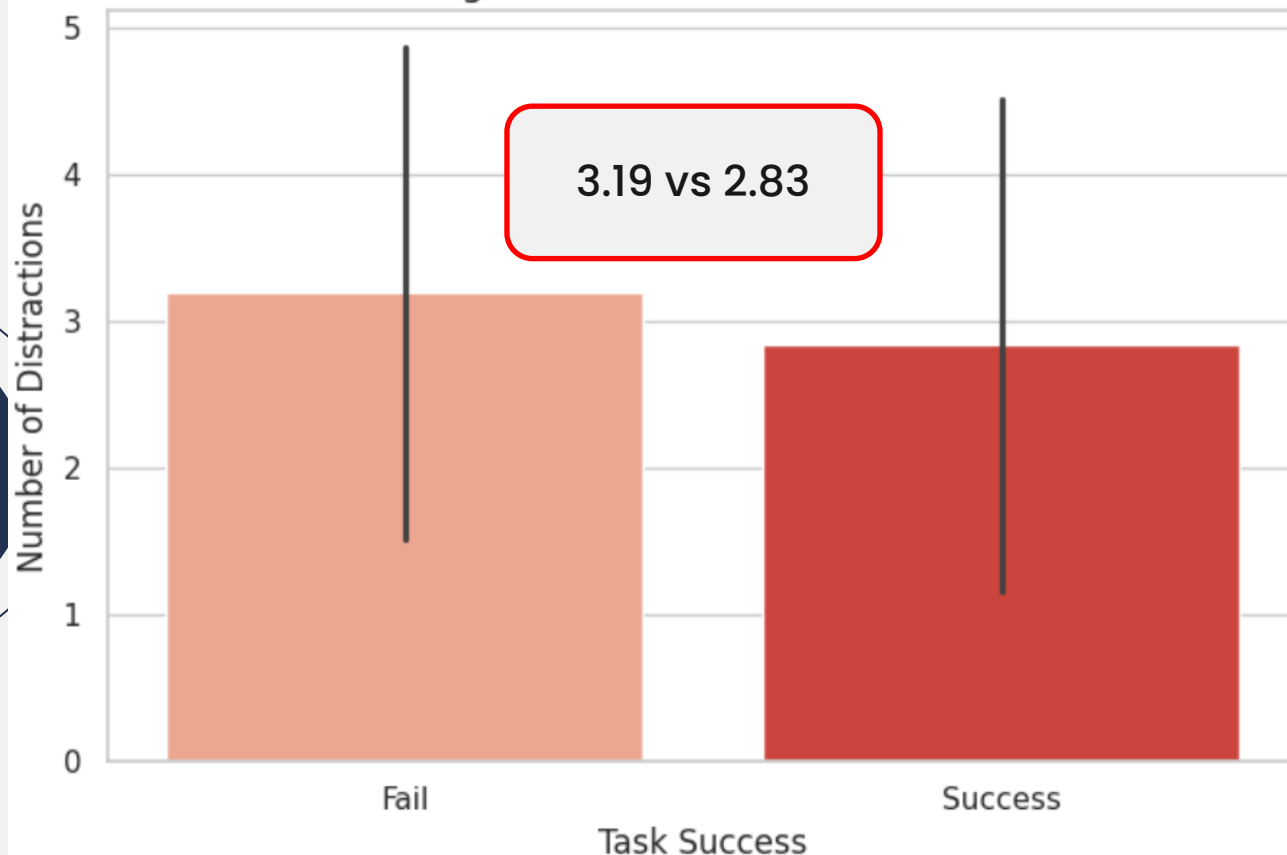
Sleep Hours
=
% Success

- Sleep Hours
- Coffee Intake
- Distractions
- Hours Coding
- AI Usage Hours
- **Task Success** – whether the daily productivity goal was achieved

More coffee,
more success!

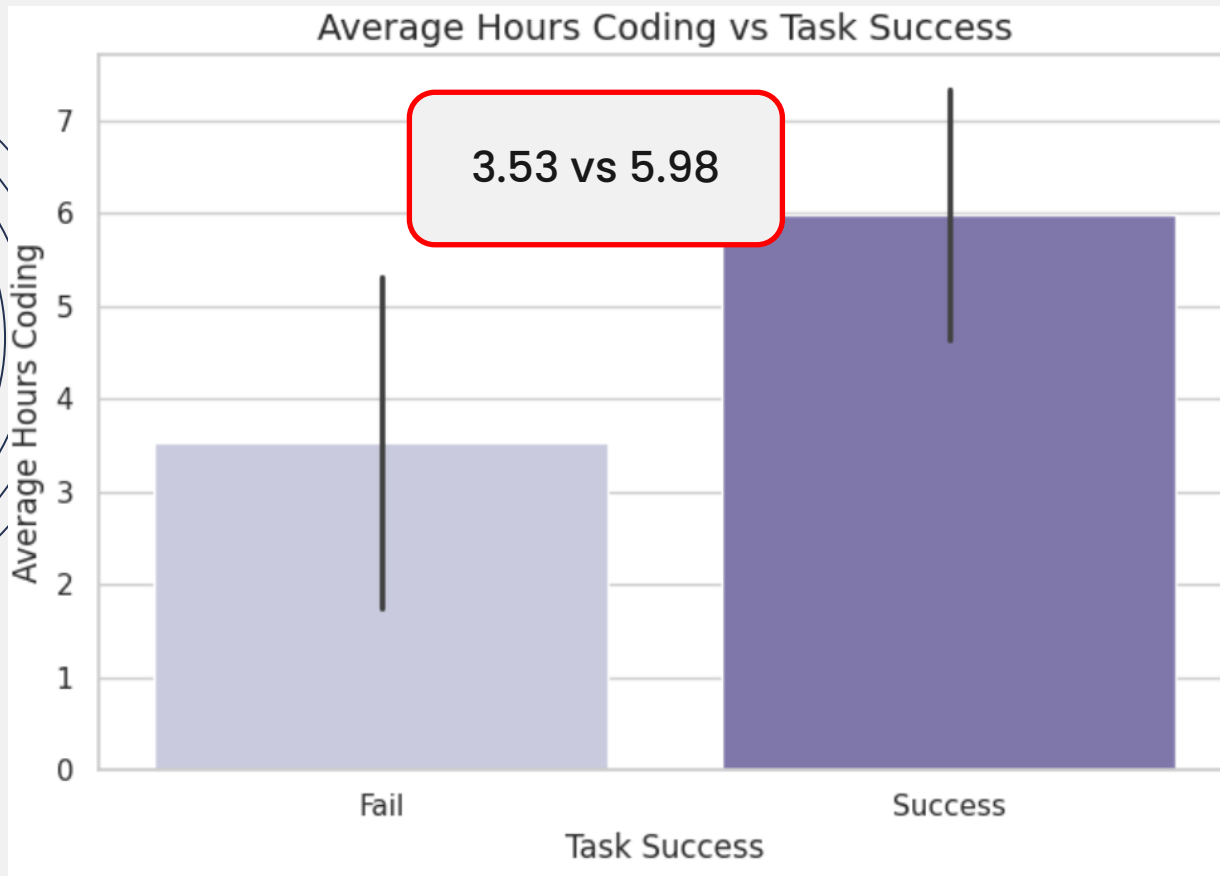


Average Distractions vs Task Success



- Sleep Hours
- Coffee Intake
- Distractions
- Hours Coding
- AI Usage Hours
- Task Success - whether the daily productivity goal was achieved

Less distractions
→ success



- Sleep Hours
- Coffee Intake
- Distractions
- Hours Coding
- AI Usage Hours
- **Task Success** – whether the daily productivity goal was achieved

Likely that more hours coding -> higher success

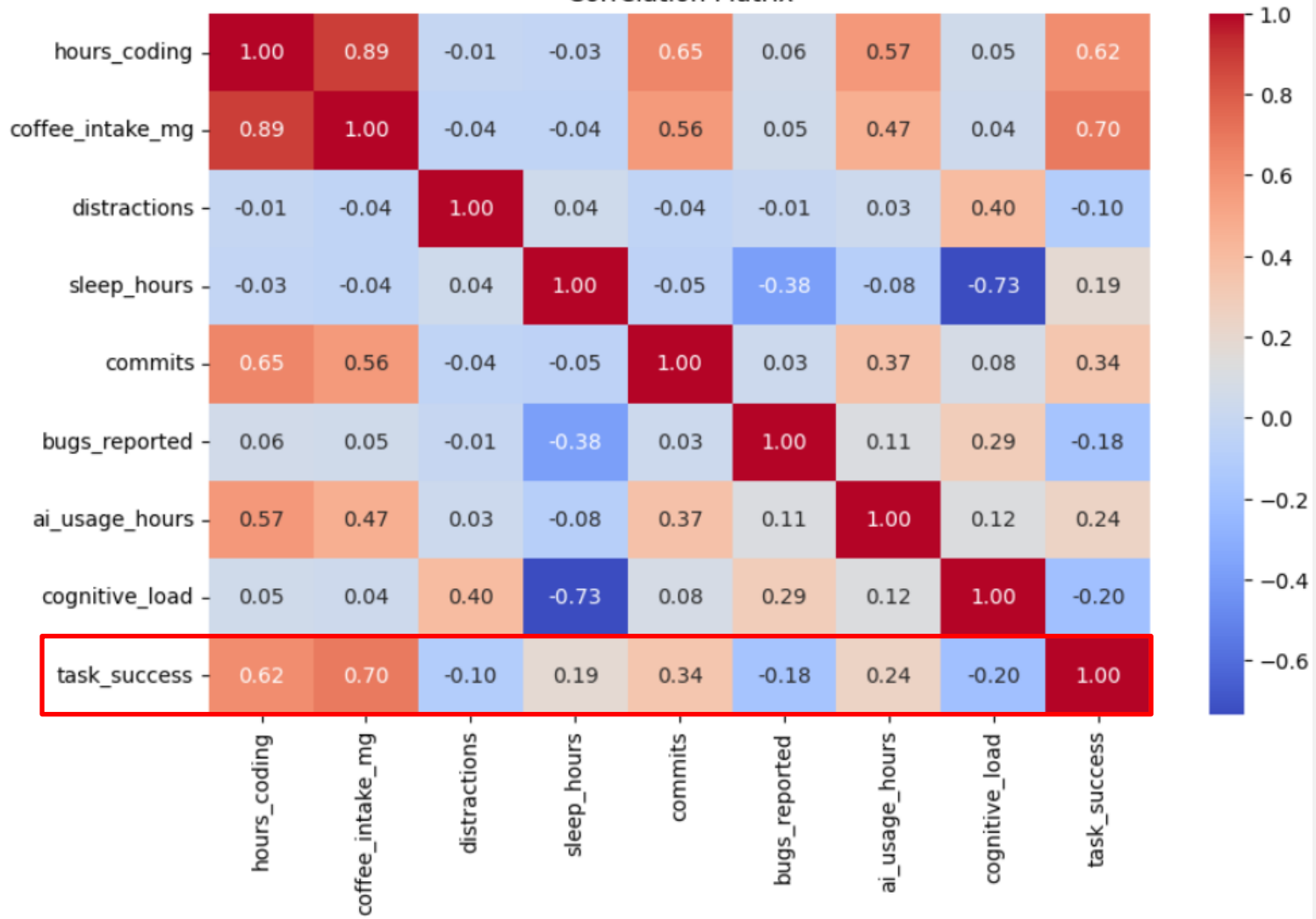
Average AI Usage vs Task Success

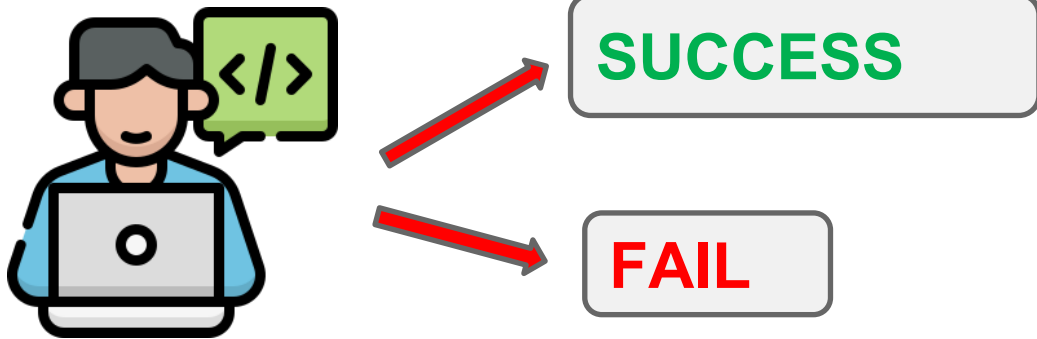


- Sleep Hours
- Coffee Intake
- Distractions
- Hours Coding
- AI Usage Hours
- Task Success – whether the daily productivity goal was achieved

Higher AI
usage likely
beneficial

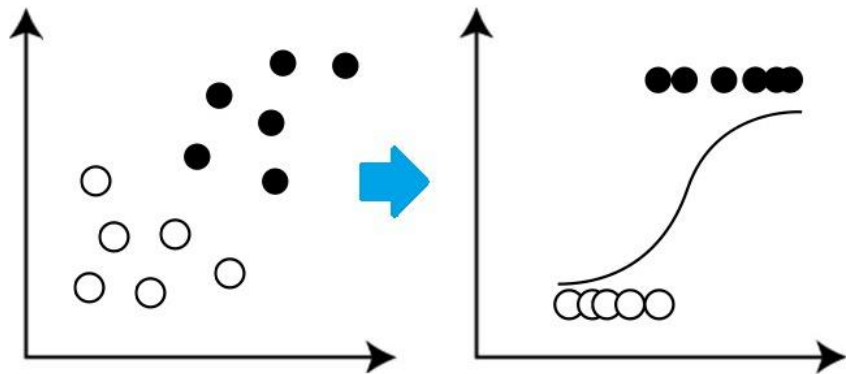
Correlation Matrix



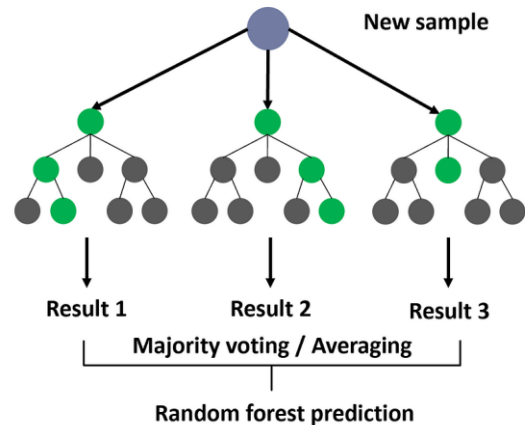


Supervised Learning Models

Logistic Regression



Random Forest



RESULTS

Confusion Matrix - Logistic Regression

Actual	Predicted	
	Predicted: Fail	Predicted: Success
Actual: Fail	28	6
Actual: Success	9	57

$$(28 + 57) / (28 + 6 + 9 + 57) = 85\%$$

Confusion Matrix - Logistic Regression (Training Data)

Actual	Predicted	
	Predicted: Fail	Predicted: Success
Actual: Fail	137	26
Actual: Success	18	219

$$(137 + 219) / (137 + 26 + 18 + 219) = 89\%$$

Random Forest

Confusion Matrix - Random Forest (Test Data)

Actual	Predicted	
	Predicted: Fail	Predicted: Success
Actual: Fail	34	0
Actual: Success	0	66

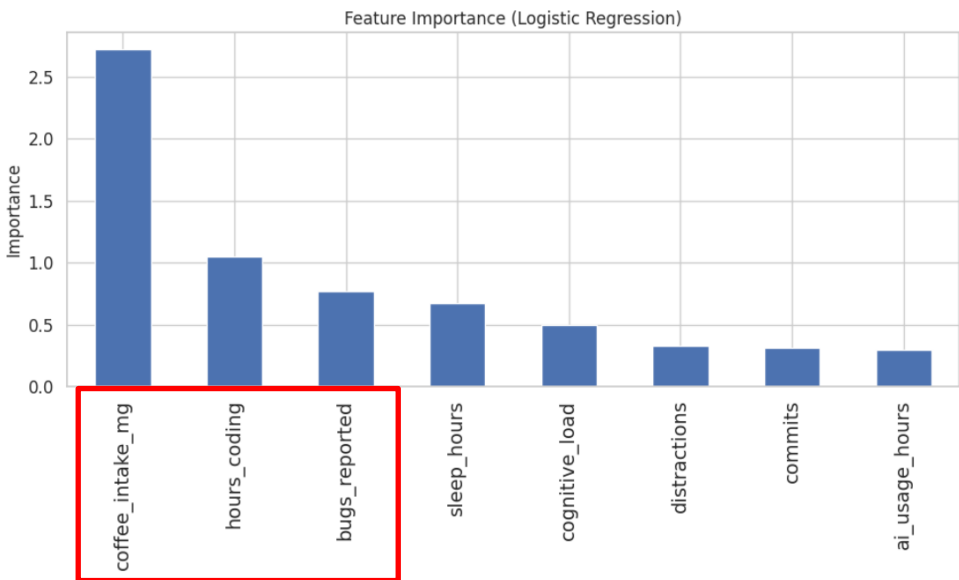
100%

Confusion Matrix - Random Forest (Training Data)

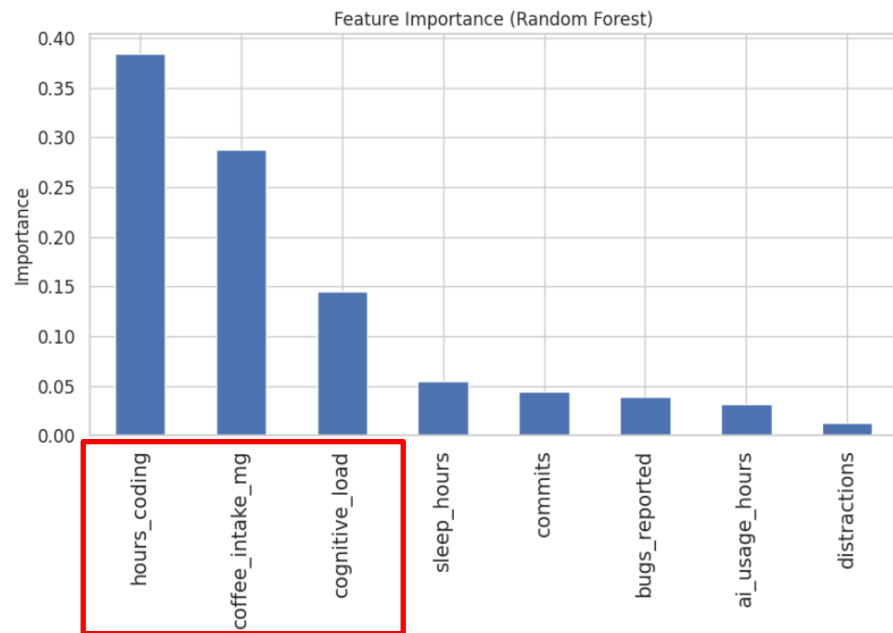
Actual	Predicted	
	Predicted: Fail	Predicted: Success
Actual: Fail	163	0
Actual: Success	0	237

100%

Feature Importance



Top features!



Top features!

Conclusions



1. Developer productivity is predictable
2. Random Forest outperformed
3. Success driven by behaviour
4. Productivity can be optimised with simple behavioural changes



- Sleep Hours (8%)
- Coffee Intake (59%)
- Hours Coding (69%)
- AI Usage Hours (45%)



- Distractions (11%)



Recommendations

Developers



Focused coding time

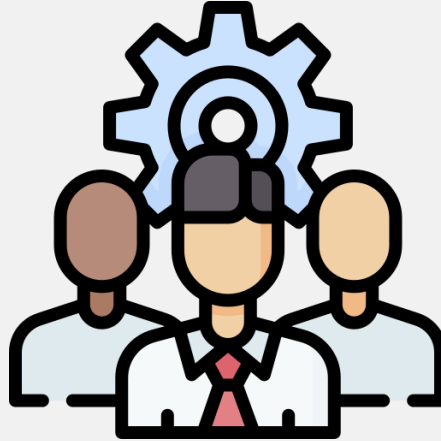
Use AI tools

Minimise distractions

Use caffeine strategically

Prioritise sleep

Team Leads



Re-evaluate meeting culture

Use predictive dashboards

Human Resources



Promote sleep, recovery

Support deep work

Onboarding

Invest in better/free coffee



Assumptions & Limitations

Assumptions:

- Developers are defining “task success” in a similar way
- No major external influences
- Each row is independent

Limitations:

- Sample size (500)
- Time-based effects
- Task complexity
- Correlation, not cause
- Basic features

Additional Data:

- More samples
- Task context
- Day of Week
- Code Quality



Thank you!
Code review

